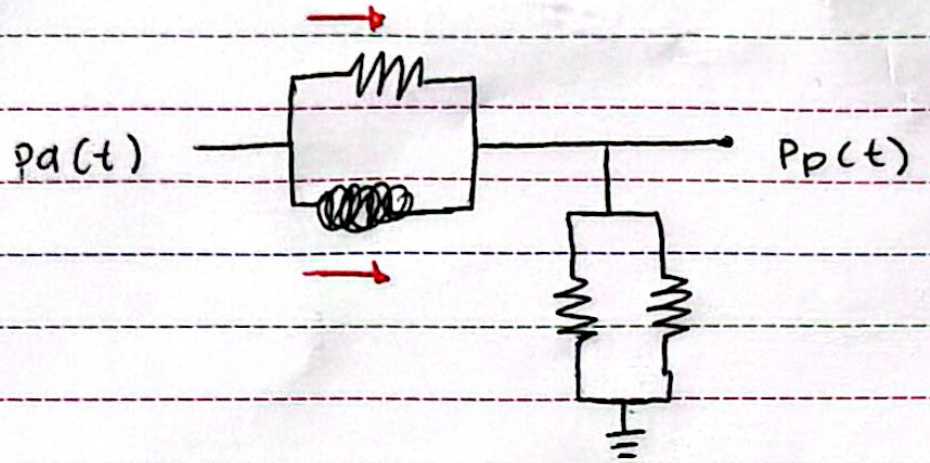
modelado de sistemas fisiológicos



$$Fa(t) = Fz(t) + F_L(t) = Fc(t) + F_R(t)$$

$$Fz(t) = Pa(t) - Pp(t) / z$$

$$F_L(t) = \frac{1}{L} \int [Pa(t) - Pp(t)]\, dt$$

$$Fc(t) = C \frac{dPp(t)}{dt}$$

$$F_R(t) = Pp(t) / R$$

TRANSFORMADA DE LAPLACE

$$Fa(s) = Fz(s) + F_L(s) = Fc(s) + F_R(s)$$

$$Fz(s) = Pa(s) - Pp(s) / z$$

$$F_L(t) = Pa(s) - Pp(s) / Ls$$

$$Fc(t) = Pp(s) / R$$

$$\frac{Pa(s) - Pp(s)}{z} + \frac{Pa(s) - Pp(s)}{Ls} = CsPp(s) + \frac{Pp(s)}{R}$$

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modelo de ecuaciones integro-diferenciales ...

$$F_2(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{P_a(t) - P_p(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt = C\frac{d P_p(t)}{dt} + \frac{P_p(t)}{R}$$

$$P_p(t)\left(\frac{1}{R} + \frac{1}{Z}\right) = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt - C\frac{dP_p(t)}{dt}\right]$$

$$P_p(t)\frac{Z+R}{R_2} = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt - C\frac{dP_p(t)}{dt}\right]$$

$$P_p(t) = \left[[P_a(t) - P_p(t)]\,dt - C\frac{dP_p(t)}{dt}\right]\frac{RZ}{Z+R}$$



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Procedimiento ...

$$\frac{Pa(s)}{2} - \frac{Pp(s)}{2} + \frac{Pa(s)}{Ls} - \frac{Pp(s)}{Ls} = Cs\,Pp(s) + \frac{Pp(s)}{R}$$

$$\left(\frac{1}{2} + \frac{1}{Ls}\right) Pa(s) = \left(Cs + \frac{1}{R} + \frac{1}{2} + \frac{1}{Ls}\right) Pp(s)$$

$$\frac{Ls+2}{L2s}\,Pa(s) = \frac{CLR2s^2 + L(2+R)s + R2}{LR2s}\,Pp(s)$$

$$\frac{Pp(s)}{Pa(s)} = \frac{\frac{Ls+2}{L2s}}{\frac{CLR2s^2 + L(2+R)s + R2}{LR2s}}$$

Resultado ...

$$\frac{Pp(s)}{Pa(s)} = \frac{LRs + R2}{CLR2s^2 + L(2+R)s + R2}$$

error en estado estacionario ...

este error se calcula cuando la entrada $Pa(s)$ es un escalon unitario

$$e(s) = \lim_{s \to 0} s\,Pa(s) \qquad = Pa(s) = 1/s$$

$$\lim_{s\to 0} s\,\frac{1}{s}\left(1-\frac{LRs+R2}{CLR2s^2+L(2+R)s+R2}\right)=0$$

$$e(t)=0v.$$

estabilidad del sistema en lazo abierto...

para analizar la estabilidad del sistema en lazo abierto se debe calcular los polos:

$$CLR2s^2+L(2+R)s+R2=0$$

$$a=CLR2$$

$$b=L(2+R)$$

$$c=R2$$

$$\lambda_{1,2}=\frac{-L(2+R)\pm\sqrt{L2(2+R)2-4CLR2-4CLR^{\wedge}22}}{2CLR2}$$

Hipotenso...

respuesta estable SUBAMORTIGUADA

$z=0.010$

$L=0.001$

$C=3.500$

$$\frac{Pp(s)}{Pa(s)}=\frac{0.001s+0.01}{3.50\times10^{-5}s2+0.00101s+0.01}$$

$\lambda_1=-14.429+8.8051i$

$\lambda_2=-14.429-8.8051i$



Normotenso ...

$Z = 0.010$

$L = 0.001$ $\quad PP(s) = \dfrac{0.001s + 0.01}{1.5 \times 10^{-5} s^2 + 0.00101s + 0.01}$

$R = 1.000$ $\quad Pa(s)$

$C = 1500$

$\lambda_1 = -12.062 \qquad \lambda_2 = -55.272$

sobreamortiguada

Hipertenso ...

$Z = 0.010$

$L = 0.001$ $\quad PP(s) = \dfrac{0.001s + 0.01}{0.00001s^2 + 0.00101s + 0.01}$

$R = 1.000$ $\quad Pa(s)$

$C = 1.000$

$\lambda_1 = -10.0 \qquad \lambda_2 = -10.9$

respuesta estable y críticamente amortiguada

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